3.2 Medical Requirements Overview

TABLE 3.2: MEDICAL REQUIREMENTS OVERVIEW

MEDB# and Title:	MEDB 1.8_1.8.1 Hearing Assessment (includes MED Volume B Sections 1.8 and 1.8.1)
Sponsor:	Medical Operations
Discipline:	Neurological Function/Acoustic
Category:	Medical Requirements (MR)
References:	ISS Medical Operations Requirements Document (MORD) SSP 50260: Medical Evaluation Documents (MED) Volume B Section 1.8 Hearing Assessment Medical Evaluation Documents (MED) Volume B Section 1.8.1 Hearing Protection
Purpose/Objectives:	To assess (and detect any changes in) crewmember hearing status preflight, on orbit, and post-flight
Measurement Parameters:	Hearing assessments, consisting of audiometric tests at specified frequencies within the range of 250 Hz – 10,000 Hz, tympanometry, and additional audiological tests as determined by an audiologist.
Deliverables:	Hearing assessment report - reviewed by Audiologist and Crew Surgeon. On-Orbit Hearing Assessments results will be sent down via data downlink within the week of on-orbit test (this is private data).
Flight Duration:	$\geq$ 30 days
Number of Flights:	All ISS Increments
Number and Type of Crew Members Required:	All Crewmembers
Other Flight Characteristics:	N/A

3.3 Preflight Training

TABLE 3.3: PREFLIGHT TRAINING

Preflight Training Activity Description:	Crewmember will be trained on the set up of the On-Orbit Hearing Assessment (OOHA) protocol, using KUDUwave application on any Station Support Computer (SSC) laptop. As detailed in MR084L, this training lesson also includes the use of hearing protection, as well as training on the use of the Acoustic Monitor.			
Schedule:	**Duration:**	**Schedule:**	**Flexibility:**	**Personnel Required:**
	1st Instance - EHS Acoustics in the Pre-Assignment Training Plan (60 minutes) 2nd Instance - Covered in the EHS Assessment (5 min of a 90-minute lesson), which is part of the assigned crew training flow	No earlier than L-18 months	N/A	Instructors, Crewmembers
Ground Support Requirements Hardware/Software	**Preflight Hardware:**	**Preflight Software:**	**Test Location:**	
	Acoustic Monitor, Space Station Computer (SSC) Laptop, Mini USB Cable, Hearing Protection Devices (Custom Prophonics and Etymotics, Foam earplugs), ANR Headset (Bose), OOHA Headset, OOHA Ear Tips, OOHA CalPod, OOHA USB Cable, AAA Batteries (for Bose QC2 headsets), AA Batteries (for Acoustic Monitor)	Certified "Supervisor" on SSC (for Acoustic Monitor) and KUDUwave software (for OOHA)	US	
Training Facilities	**Minimum Room Dimensions:**	**Number of Electrical Outlets:**	**Temperature Requirements:**	**Special Lighting:**
	Approx. 29 feet X 14 feet	2	Ambient	N/A
	Hot or Cold Running Water:	**Privacy Requirements:**	**Other:**	
	N/A	Private room, free of distraction.	One table, 6-8 chairs.	
Constraints/Special Requirements:	N/A			
Launch Delay Requirements:	Crewmember may request refresher training if desired.			
Notes:				

3.4 Preflight Activities

TABLE 3.4.1: PREFLIGHT ACTIVITIES

Preflight Activity Description:	Ear impressions of crewmember's external ear canals will be made by audiologist. Impressions will then be used in fabrication of custom ear wear for Prophonics earplugs set earwear (as well as for custom Etymotics earplugs as described in MR084L). Crewmembers will become familiar with ISS hearing protection devices (MEDB 1.8.1). This activity is done consecutively with similar procedures described in MR084L Acoustic Monitoring & Countermeasures (Table 3.4 Preflight Activities Session 1).			
Schedule:	**Duration:**	**Schedule:**	**Flexibility:**	**Personnel Required:**
	20 minutes (represents the time for ear impressions) 30 minutes in total when performed in conjunction with MR084L	No earlier than L-30 months	N/A	Crewmembers, Audiologist
Ground Support Requirements Hardware/Software	**Preflight Hardware:**	**Preflight Software:**	**Test Location:**	
	Audiologist's materials for fabricating silicone impressions of external ear canals	N/A	U.S./JSC Clinic	
Testing Facilities	**Minimum Room Dimensions:**	**Number of Electrical Outlets:**	**Temperature Requirements:**	**Special Lighting:**
	8' x 10'	None	Ambient	N/A
	Hot or Cold Running Water:	**Privacy Requirements:**	**Vibration/Acoustic Isolation:**	**Other:**
	Sink for handwashing	Private room free of distractions	N/A	Table, 2 chairs, Otoscope
Constraints/Special Requirements:				
Launch Delay Requirements:				
Notes:				
Data Delivery	An entry in the crewmember's Electronic Medical Record will document that ear impressions were made and that familiarization training was completed			

TABLE 3.4.2: PREFLIGHT ACTIVITIES

Preflight Activity Description: **Schedule:**	Crewmember will be fit-checked to ensure proper fit of the custom-made ear wear (Prophonics Earplugs Set and Etymotics earplugs as described in MR084L), be familiarized with correct method of inserting custom-made ear wear, fit for the proper OOHA Ear Tip size, perform an initial ground-based Functional Check and baseline OOHA with the KUDUwave software, and be tested with conventional audiometry and tympanometry to obtain baseline hearing data prior to flight. The Functional (listening) Check of the Prophonics ear wear will be conducted to confirm that the crewmember finds the custom-made ear wear to be comfortable and confirm that the crewmember can hear an audible signal in each ear. The audiologist will measure the amount of noise attenuation provided by the Prophonics when worn by the crew member. To serve as a reference for the baseline OOHA (with results that vary from conventional audiometric results, due to use of the KUDUwave technique, and the acoustic output of the SSC laptop), conventional audiometry will be conducted, using an ANSI-calibrated audiometer and earphones, immediately after the baseline OOHA (MEDB 1.8.1). This activity is done consecutively with similar procedures described in MR084L Acoustic Monitoring & Countermeasures (Table 3.4 Preflight Activities Session 2).			
	Duration:	**Schedule:**	**Flexibility:**	**Personnel Required:**
	55 minutes (represents time focused on OOHA familiarization and test, and Hearing Assessment) 75 minutes in total when performed in conjunction with MR084L	L-20/5 months Fit check must occur at least 30 days after ear impressions	N/A	Crewmember, Audiologist, Hardware Engineer
Ground Support Requirements Hardware/Software	**Preflight Hardware:**	**Preflight Software:**	**Test Location:**	
	SSC laptop, Prophonics Molded Earplugs Set, Etymotics Earplugs, OOHA Headset, OOHA Ear Tips, CalPod, Air-Conduction Audiometer, Tympanometer, Sound-treated Audiometric Booth	KUDUwave software	U.S./JSC Clinic	
Testing Facilities	**Minimum Room Dimensions:**	**Number of Electrical Outlets:**	**Temperature Requirements:**	**Special Lighting:**
	8’ x 10’ (for KUDUwave)	Two	Ambient	N/A
	Hot or Cold Running Water:	**Privacy Requirements:**	**Vibration/Acoustic Isolation:**	**Other:**
	N/A	Private room free of distractions	Low noise levels	One table, 2 chairs, Otoscope
Constraints/Special Requirements:	Crewmember should not be exposed to loud noises (greater than 85dBA or weapons firing) for at least 16 hours prior to conventional hearing assessment and baseline OOHA.			
Launch Delay Requirements:				
Notes:				
Data Delivery	A copy of the baseline OOHA and reference audiometry and tympanometry data will also be included in the crewmember's Electronic Medical Record.			

TABLE 3.4.3: PREFLIGHT ACTIVITIES

Preflight Activity Description:	Crewmember will be tested using conventional audiometry and tympanometry during their last rotation L-90/30 (no earlier than L-90±15days).			
Schedule:	**Duration:**	**Schedule:**	**Flexibility:**	**Personnel Required:**
	Audiometry: 20 minutes	During last rotation L-90/30 (no earlier than L-90±15days)	Flexibility in schedule so that test is performed at the JSC Flight Medicine Clinic.	Crewmember, Audiologist
Ground Support Requirements Hardware/Software	**Preflight Hardware:**	**Preflight Software:**	**Test Location:**	
	Sound-treated Audiometric Booth Audiometry equipment Note: Hardware provided by JSC Clinic	Audiometry software provided by JSC Clinic	U.S./JSC Clinic	
Testing Facilities	**Minimum Room Dimensions:**	**Number of Electrical Outlets:**	**Temperature Requirements:**	**Special Lighting:**
	Approximately 4 ft. X 5 ft.	2	Normal range	N/A
	Hot or Cold Running Water:	**Privacy Requirements:**	**Vibration/Acoustic Isolation:**	**Other:**
	N/A	A sound-treated audiometric booth is provided by the JSC Clinic	Acoustic isolation is necessary	N/A
Constraints/Special Requirements:	• Test will be conducted in the U.S. at the JSC Flight Medicine Clinic. • Crewmember should not be exposed to loud noises (greater than 85 dBA or weapons firing) for at least 16 hours prior to hearing assessment.			
Launch Delay Requirements:	N/A			
Notes:				
Data Delivery	Results from the hearing assessment will be included in the crewmember's Electronic Medical Record.			

3.5 In-Flight Activities

TABLE 3.5.1: IN-FLIGHT ACTIVITIES

In-Flight Activity Description:	The OOHA will assess changes in crew hearing on-orbit. The crewmember will be scheduled to perform an OOHA using the KUDUwave software on any on-orbit SSC laptop on or before the first 21 days of flight and approximately mid-mission.				
	Activity:	**Duration:**	**Schedule:**	**Flexibility:**	**Personnel Required:**
Schedule:	OOHA using KUDUwave	Unstow: 5 minutes per day (Single/1st CM) Calibration Check: 5 minutes per day Hearing Assessment: 20 minutes per CM Stow: 5 minutes per day (Single/Last CM)	On or before FD 21, and approximately mid-mission, with supplemental OOHA's as dictated by noise environment or crewmember concerns. <u>If mission length planned/ extended > 6 months</u> On or before FD 21, then every 3 – 4 months. OOHAs will be scheduled within 24 hours following acoustic monitor measurements.	• First test may occur on or before FD21. • May begin as early as 3 hours before pre-sleep on day of acoustic monitor measurements and as late as 24 hours after end of measurements. • Latitude to allow scheduling efficiency across multiple expedition crews	Crewmembers
Procedures:	SODF: Med Ops book				
Constraints / Special Requirements:	• Deployed location for this test will be at any SSC laptop on-orbit, in a quiet location. Crewmember can use the same location for uniform conditions among all test results. • OOHA scheduled within 24 hours following acoustic monitor measurements. • Crew members performing OOHA's on the same day should be scheduled consecutively. OOHA Hardware (per OCAD OOHAHEADSET-001) hardware must be stored in sealed bag when not in use, scheduling consecutively is to achieve crew time savings. • If the OOHA hardware is fully stored between CMs performing KUDUwave on the same day the unstow and stow activities will need to be repeated.				
Photo / TV Requirements:	N/A				
Cold Stowage Requirements:	N/A				
Mission Extension Requirements:	N/A				
Landing Wave-Off Requirements:	N/A				
Data Delivery	Results from the OOHA will be saved to an on-board CHeCS data folder on the ISS OPS LAN file server and downloaded to the ground within a week from the test. The Audiologist will deliver a report to the FS within 3 days after receiving data. If a significant threshold shift is detected, the crew will be referred back to the in-flight procedures for corrective action.				

In-Flight Activities, (cont.)

TABLE 3.5.2: IN-FLIGHT HARDWARE

Hardware/Software Name
SSC
KUDUwave software
OOHA Headset
OOHA Ear Tips
OOHA USB Cable
OOHA CalPod

3.6 Postflight Activities

TABLE 3.6: POSTFLIGHT ACTIVITIES

Postflight Activity Description:	Crewmember will be tested using conventional audiometry and tympanometry at R+3 days. When possible, these tests will be combined with the postflight physical exam on the same day (MEDB 1.1 Long duration Pre/Postflight Physical Exam).			
Schedule:	**Duration:**	**Schedule:**	**Flexibility:**	**Personnel Required:**
	Audiometry: 20 minutes	R+3 days (If abnormal, R+10/14, R+60)	Flexibility in schedule so that test is performed at the JSC Flight Medicine Clinic.	Crewmember, Audiologist
Ground Support Requirements Hardware/Software	**Postflight Hardware:**	**Postflight Software:**	**Test Location:**	
	Sound-treated audiometric booth, Audiometry equipment Note: Hardware provided by JSC Clinic	Audiometry software provided by JSC Clinic	U.S./JSC Clinic	
Testing Facilities	**Minimum Room Dimensions:**	**Number of Electrical Outlets:**	**Temperature Requirements:**	**Special Lighting:**
	Approximately 4 ft. X 5 ft.	2	Normal range	N/A
	Hot or Cold Running Water:	**Privacy Requirements:**	**Vibration/Acoustic Isolation:**	**Other:**
	N/A	A sound-treated audiometric booth is provided by the JSC Clinic	Acoustic isolation is necessary	N/A
Constraints/Special Requirements:	• Crewmember should not be exposed to loud noises (greater than 85 dBA or weapons firing) for at least 16 hours prior to audiometry.			
Early Destow / Early Return:	N/A			
Notes:	When possible, this test is combined with the post-flight physical (MEDB 1.1 Long Duration Pre/Postflight Physical Exam) on R+3 days.			
Data Delivery	Results from the audiometry will be included in the crewmember's medical report and Electronic Medical Record.			

TABLE 3.7: SUMMARY SCHEDULE

ACTIVITY	DURATION	SCHEDULE	FLEXIBILITY	PERSONNEL REQUIRED	CONSTRAINTS
Preflight Training					
1st Instance - EHS Acoustics in the Pre-Assignment Training Plan 2nd Instance - Covered in the EHS Assessment (5 min of a 90-minute lesson), which is part of the assigned crew training flow	1st Instance - 60 minutes 2nd Instance – 5 minutes	No earlier than L-18 months	N/A	Instructors, Crewmembers	None
Preflight Activities					
Ear impressions	20 minutes (represents the time for ear impressions) 30 minutes in total when performed in conjunction with MR084L	NET L-30 months	N/A	Crewmembers, Audiologist	Crewmember's ear canals must be free of earwax, which alter the shape of the actual ear canal. Excessive earwax, if seen, must be removed prior to ear impression.
Fit/ Function Check, OOHA, Hearing Assessment	55 minutes (75 minutes in total when performed in conjunction with MR084L)	L-20/5 months	N/A	Crewmember, Audiologist, Hardware Engineer	Crewmember should not be exposed to loud noises (greater than 85 dBA or weapons firing) for at least 16 hours prior to conventional audiometry and baseline OOHA.
Hearing Assessment	20 minutes	During last rotation (no earlier than L-90/30 days)	Flexibility in schedule so that test is performed at the JSC Flight Medicine Clinic.	Crewmember, Audiologist	Test will be conducted in the U.S. at the JSC Flight Medicine Clinic. Crewmember should not be exposed to loud noises (greater than 85 dBA or weapons firing) for at least 16 hours prior to audiometry.

MEDB 1.8_1.8.1 Hearing Assessment

ACTIVITY	DURATION	SCHEDULE	FLEXIBILITY	PERSONNEL REQUIRED	CONSTRAINTS
In-Flight					
OOHA with KUDUwave	Unstow: 5 minutes per day (Single/1st CM) Calibration Check: 5 minutes per day Hearing Assessment: 20 minutes per CM Stow: 5 minutes per day (Single/Last CM)	On or before FD 21, and approximately mid-mission, with supplemental OOHA's as dictated by noise environment or crewmember concerns. If mission length planned/ extended > 6 months On or before FD 21, then every 3 – 4 months. OOHAs will scheduled within 24 hours following Acoustic Monitor – Crew Worn measurements.	• First test may occur on or before FD21 • Latitude to allow scheduling efficiency across multiple expedition crews	Crewmembers	Scheduled within 24 hours following acoustic monitor measurements.
Post-Flight					
Hearing Assessment	20 minutes	R+3 days (If abnormal, R+10/14, R+60)	Flexibility in schedule so that test is performed at the JSC Flight Medicine Clinic.	Audiologist, Crewmember	Upon return to the U.S. test will be conducted at the JSC Flight Medicine Clinic. Crewmember should not be exposed to loud noises (greater than 85 dBA or weapons firing) for at least 16 hours prior to audiometry.